

需再請蔡老師微調:

1. 進行 排除個案的基本data 分析

目前還缺 2個項目的 P value (所屬機構區域分布的差異、所屬機構規模分布的差異)

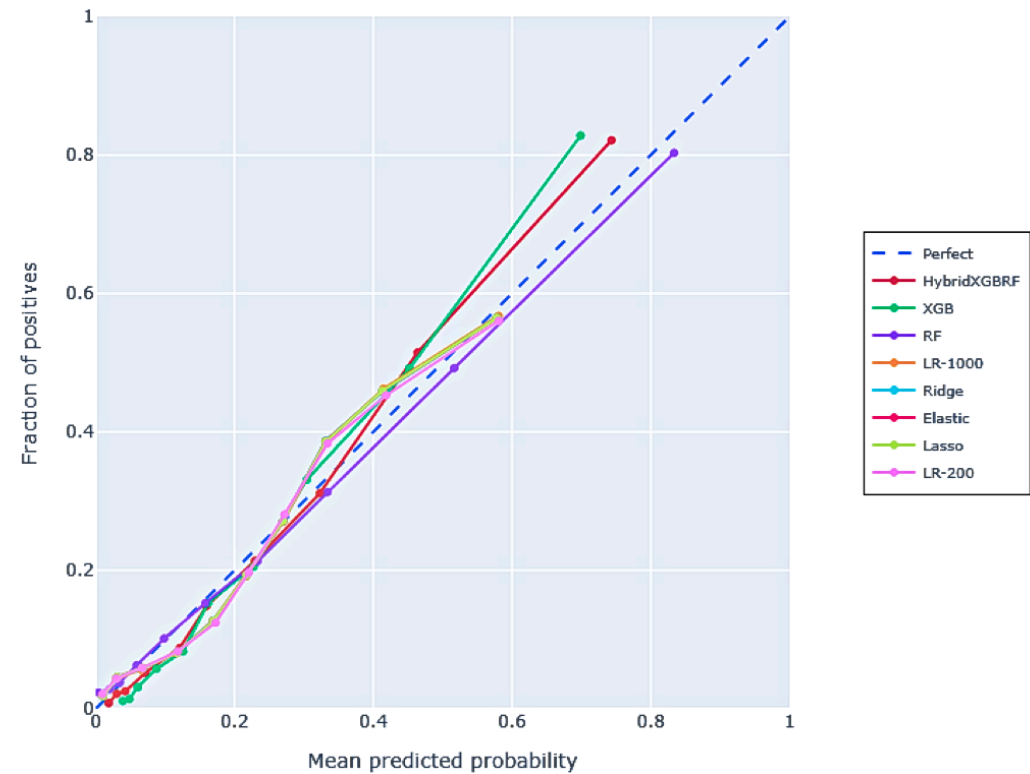
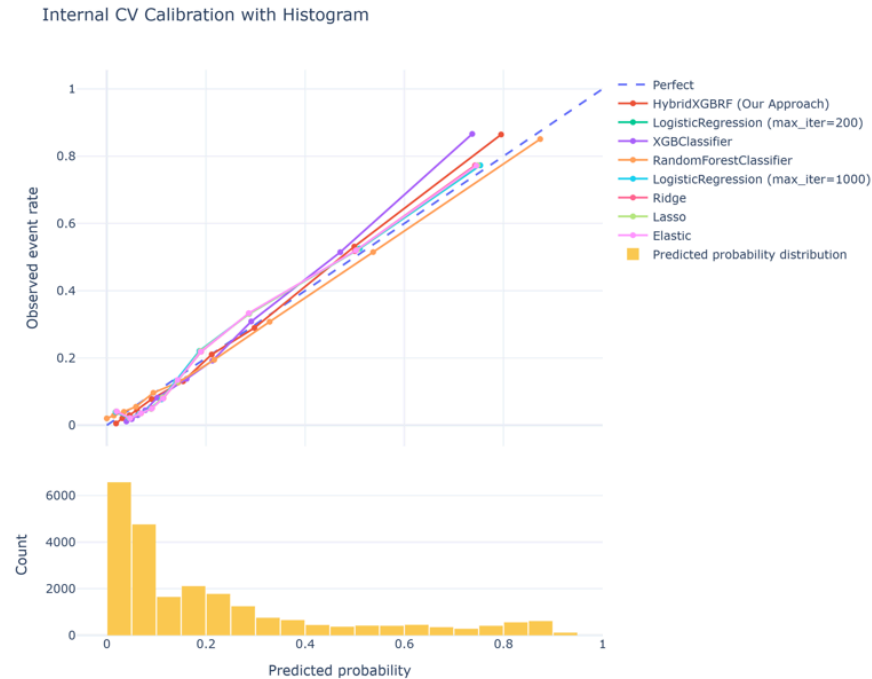
Characteristic	Development cohort (N = 23,901)	%	Excluded subjects (Unable to ascertain 6-month mortality outcome) (n = 19,756)	%	P
Institutional regions					
North	2245	9.39%	2112	10.69%	
Central	4754	19.89%	4019	20.34%	
South	16,603	69.46%	13447	68.07%	
East	299	1.25%	174	0.88%	
Institution scale					
Small (<50 beds)	311	63.08%	363	61.95%	
Medium (50-150 beds)	148	30.02%	182	31.06%	
Large (>150 beds)	34	6.90%	41	7.00%	

2. 補做 distribution histogram of the predicted probabilities in the calibration plots

想要請老師微調圖案：

(1) 原圖(下左圖) 希望調整 如同投稿時的圖案, **X、Y軸標題文字與刻度 + 右側圖例的順序 與內文一致** (如下右圖)

因為這邊回復委員意見後, 需要重新將稿件抽換成含直方圖的 calibration plots, 但新圖的XY軸標題文字與圖例的呈現順序不同。



3.因為我們的主表 Table 3已經是新的數字，須請老師協助抽換 後面的圖片。

原始 Table 3

Model metrics	Model ¹ a	Model ² b	Model ³ c	Model ⁴ d	Model ⁵ e	Model ^{6f}	Model ^{7g}	Model ^{8h}
AUROC (95%CI)	0.88 (0.86–0.89)	0.87 (0.86–0.89)	0.86 (0.84–0.88)	0.80 (0.78–0.81)	0.80 (0.78–0.81)	0.80 (0.78–0.81)	0.80 (0.78–0.81)	0.80 (0.78–0.81)
Precision	0.77	0.79	0.73	0.58	0.58	0.58	0.58	0.58
Recall	0.45	0.46	0.49	0.21	0.20	0.20	0.20	0.21
Sensitivity	0.45	0.46	0.49	0.21	0.20	0.20	0.20	0.21
Specificity	0.96	0.97	0.95	0.96	0.96	0.96	0.96	0.96
F ₁ Score	0.57	0.58	0.58	0.30	0.30	0.30	0.30	0.31
Accuracy, Mean (SD)	0.85 (0.005)	0.85 (0.003)	0.85 (0.005)	0.79 (0.005)	0.79 (0.005)	0.79 (0.005)	0.79 (0.004)	0.79 (0.003)
Brier Score	0.11	0.11	0.11	0.14	0.14	0.14	0.14	0.14

Table 3. Performance of the different predictive models

Model metrics	Model 1a	Model 2b	Model 3c	Model 4d	Model 5e	Model 6f	Model 7g	Model 8h
AUROC (95%CI)	0.89 (0.88–0.89)	0.88 (0.88–0.89)	0.87 (0.87–0.88)	0.86 (0.85–0.86)	0.86 (0.85–0.86)	0.86 (0.85–0.86)	0.86 (0.85–0.86)	0.86 (0.85–0.86)
Precision (95%CI)	0.79 (0.78–0.8)	0.79 (0.78–0.8)	0.75 (0.74–0.76)	0.7 (0.69–0.71)	0.7 (0.69–0.72)	0.7 (0.69–0.71)	0.7 (0.69–0.71)	0.7 (0.69–0.71)
Sensitivity/Recall (95%CI)	0.54 (0.52–0.55)	0.51 (0.5–0.52)	0.54 (0.52–0.55)	0.49 (0.48–0.51)	0.48 (0.47–0.5)	0.48 (0.47–0.5)	0.49 (0.47–0.5)	0.49 (0.48–0.51)
Specificity	0.96	0.96	0.95	0.94	0.94	0.94	0.94	0.94
F ₁ Score (95% CI)	0.64 (0.63–0.65)	0.62 (0.61–0.63)	0.62 (0.61–0.64)	0.58 (0.56–0.59)	0.57 (0.56–0.59)	0.57 (0.56–0.59)	0.57 (0.56–0.59)	0.58 (0.56–0.59)
Accuracy SD	0.87 (0.002)	0.86 (0.002)	0.86 (0.002)	0.84 (0.002)	0.84 (0.002)	0.84 (0.002)	0.84 (0.002)	0.84 (0.002)
Brier Score (95%CI)	0.1 (0.1–0.1)	0.1 (0.1–0.11)	0.11 (0.1–0.11)	0.12 (0.11–0.12)	0.12 (0.11–0.12)	0.12 (0.11–0.12)	0.12 (0.11–0.12)	0.12 (0.11–0.12)

因為許多model的數值都改變，例如

^dLogistic Regression (max_iter=1000).

^eRidge.

^fElastic.

^gLasso.

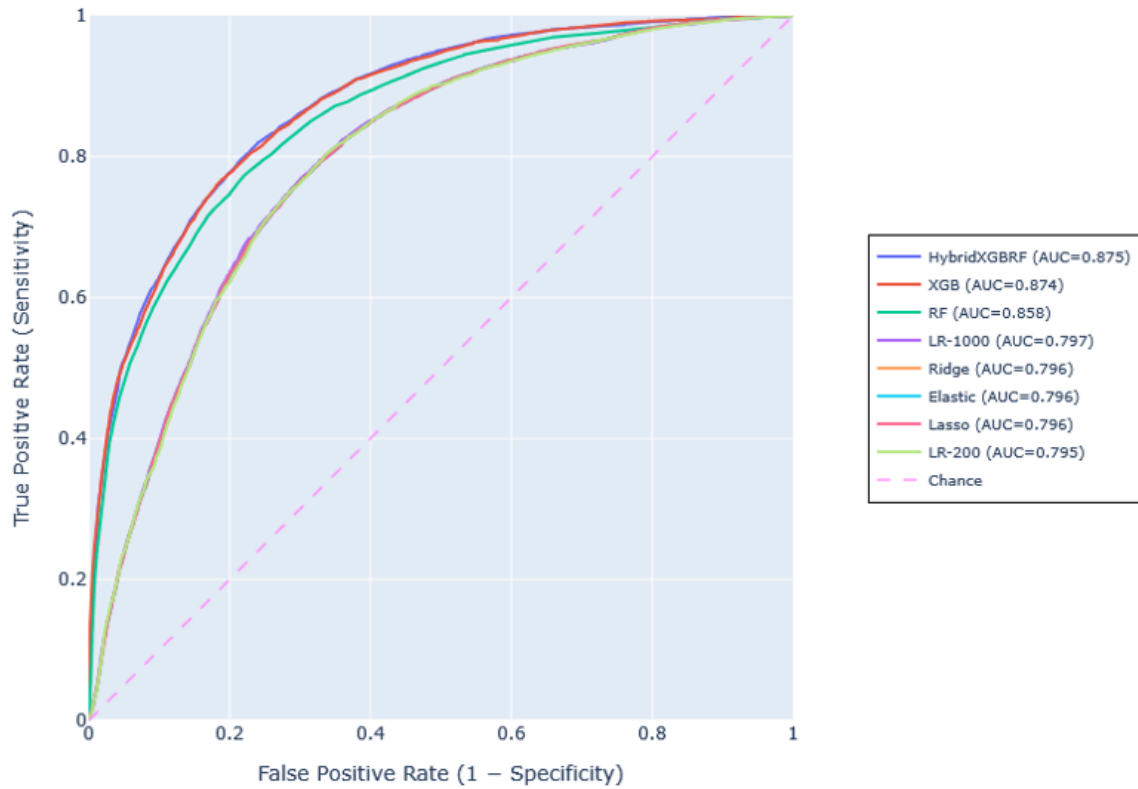
^hLogistic Regression (max_iter=200). 原本AUROC皆大約0.80，調整後為0.86 與後面的圖表變成不一致。

因為我們的主表 Table 3已經是新的數字,

須請老師協助抽換 **Figure 1 Receiver operating characteristic curve comparison** 以及 **Confusion matrix** (如下圖)

Figure 1 Receiver operating characteristic curve comparison

請老師此圖協助 X、Y軸長比例接近(1:1), X、Y軸標題名稱需與原圖(下圖)一致, 並請右側圖例順序與主文一致(如下圖)

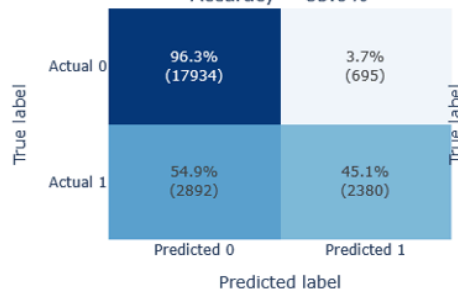


Confusion matrix (則是希望重作的圖與投稿一致, 如下圖)

Confusion matrix comparisons between machine learning classifiers (Training CV)

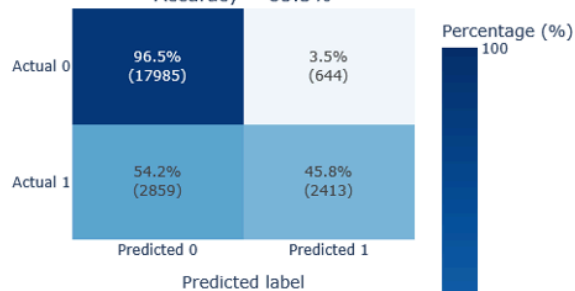
A. HybridXGBRF (Our Approach)

Accuracy = 85.0%



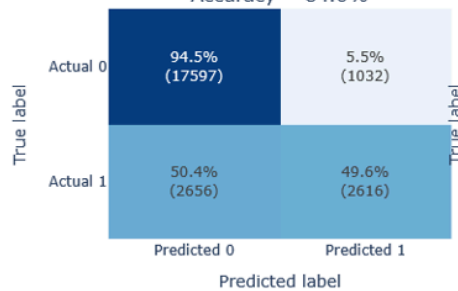
B. XGBClassifier

Accuracy = 85.3%



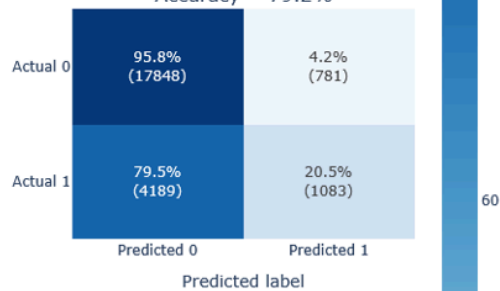
C. RandomForestClassifier

Accuracy = 84.6%



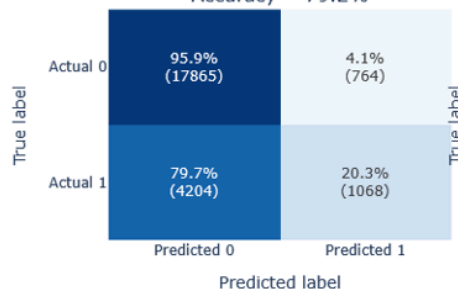
D. LogisticRegression (max_iter=1000)

Accuracy = 79.2%



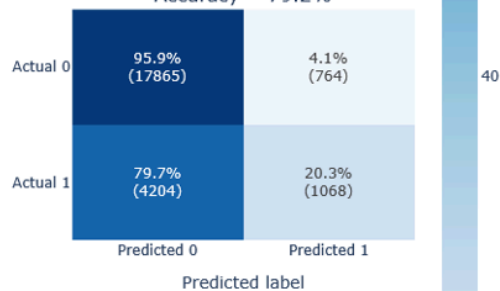
E. Ridge

Accuracy = 79.2%



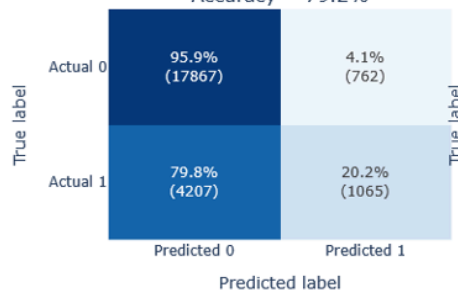
F. Elastic

Accuracy = 79.2%



G. Lasso

Accuracy = 79.2%



H. LogisticRegression (max_iter=200)

Accuracy = 79.2%

